**Fact File continue**



At first glance, it would appear that swimming in the ocean is far more dangerous than swimming in a lake. Crashing waves and riptides can easily sweep beachgoers to their doom. But shockingly enough, about 90 percent of drowning cases occur in freshwater. The reason involves a bit of chemistry. Fresh water is more similar in composition to our own blood than salt water. When it is inhaled into the lungs, it passes into the bloodstream through osmosis. When the blood is so radically diluted, cells burst, leading to organ failure. The entire process takes two to three minutes.

Ocean water contains far more salt than human blood. When it is aspirated, the body attempts to regulate itself by transferring water into the lungs, thickening the blood. It takes considerably longer to kill a person, between 8 to 10 minutes, allowing a much greater chance of rescue.

Males are especially at risk of drowning, with twice the overall mortality rate of females. Studies suggest that the higher drowning rates among males are due to increased exposure to water and riskier behavior such as swimming alone, drinking alcohol before swimming alone and boating. Drowning accounts for 75% of deaths in flood disasters.

### Prevention

There are many actions to prevent drowning. Installing barriers (e.g. covering wells, using doorway barriers and playpens, fencing swimming pools etc.) to control access to water hazards.

Community-based, supervised child care for pre-school children can reduce drowning risk. Teaching school-age children basic swimming, water safety and safe rescue skills is another approach.

Setting and enforcing safe boating, shipping and ferry regulations is an important part of improving safety on the water and preventing drowning. Building resilience to flooding and managing flood risks through better disaster preparedness planning, land use planning, and early warning systems can prevent drowning during flood disasters.

## 8.3.3 Fire Accident

Massive forest fires may start in hot and droughty weather as a result of lightning, and human carelessness or from other causal factors. Fires can lead to the destruction of buildings, wooden bridges and poles, power, transmission and telecommunication lines, warehouses of containing oil products and other fuel. It causes injury to people and animals.

**Students' activity**

**Mock Drill:** To escape a fire, **stop, drop, and roll.** In case your clothes burn, stop running, drop on the floor and roll to stop fire spreading.

During droughts or windy weather, fire may destroy low vegetation and trees. The spreading speed of low fire is 1-3 m/sec and high fire may reach up to 100m/sec.



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